Operating System in Banking Systems

CSA0407 - OPERATING SYSTEMS

B.Archana-192425325

Introduction

Operating System in Banking:

- A banking system handles thousands of transactions every day.
- **Examples:**depositing money, withdrawing money, and transferring money.
- An Operating System (OS) manages the computer's resources.
- It helps the banking system work safely and efficiently.

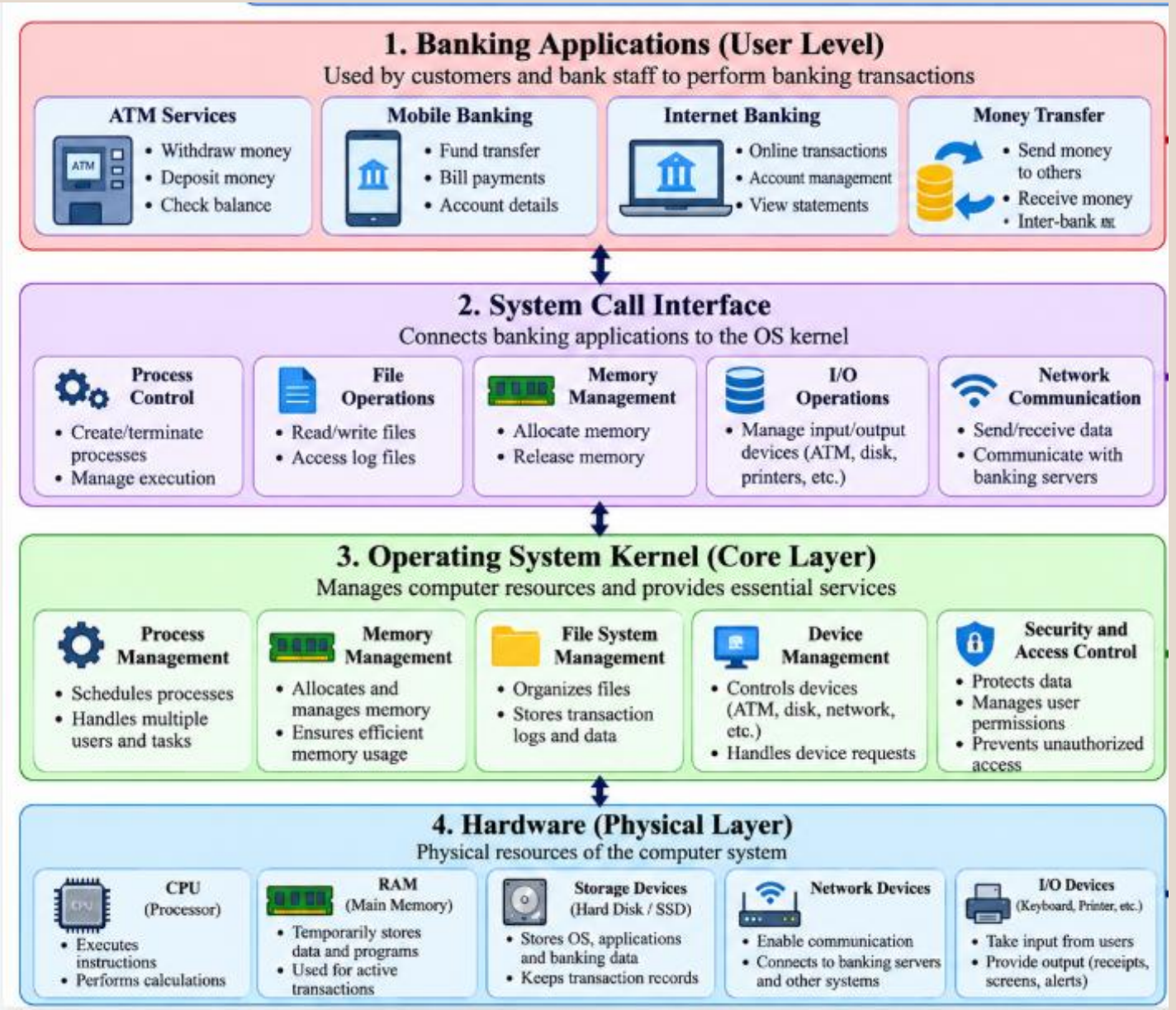
Example: When many customers use ATMs at the same time, the OS helps manage their requests.

Types of Operating Systems

- **Multiprocessing OS:** Uses multiple CPU cores to perform tasks faster.
- **Multiuser OS:** Allows many users to access the system at the same time.
- **Multitasking OS**: Handles many tasks at once.
- **Distributed OS / Distributed Computing**: Helps multiple connected computers work together.

Example: Many customers can check balances and transfer money simultaneously.

OS Structure Used



Role of System Calls

System calls allow banking applications to request services from the Operating System.

- **Process control**: Manages running tasks.
- **File management**: Reads and writes transaction records.
- **Memory management**: Provides memory for running programs.
- **Communication**: Helps computers exchange transaction requests and responses.
- **Synchronization**: Helps coordinate tasks accessing shared resources.

Example: When a customer transfers money, the application uses OS services to communicate and record information.

Transaction Processing

How a Banking Transaction Works

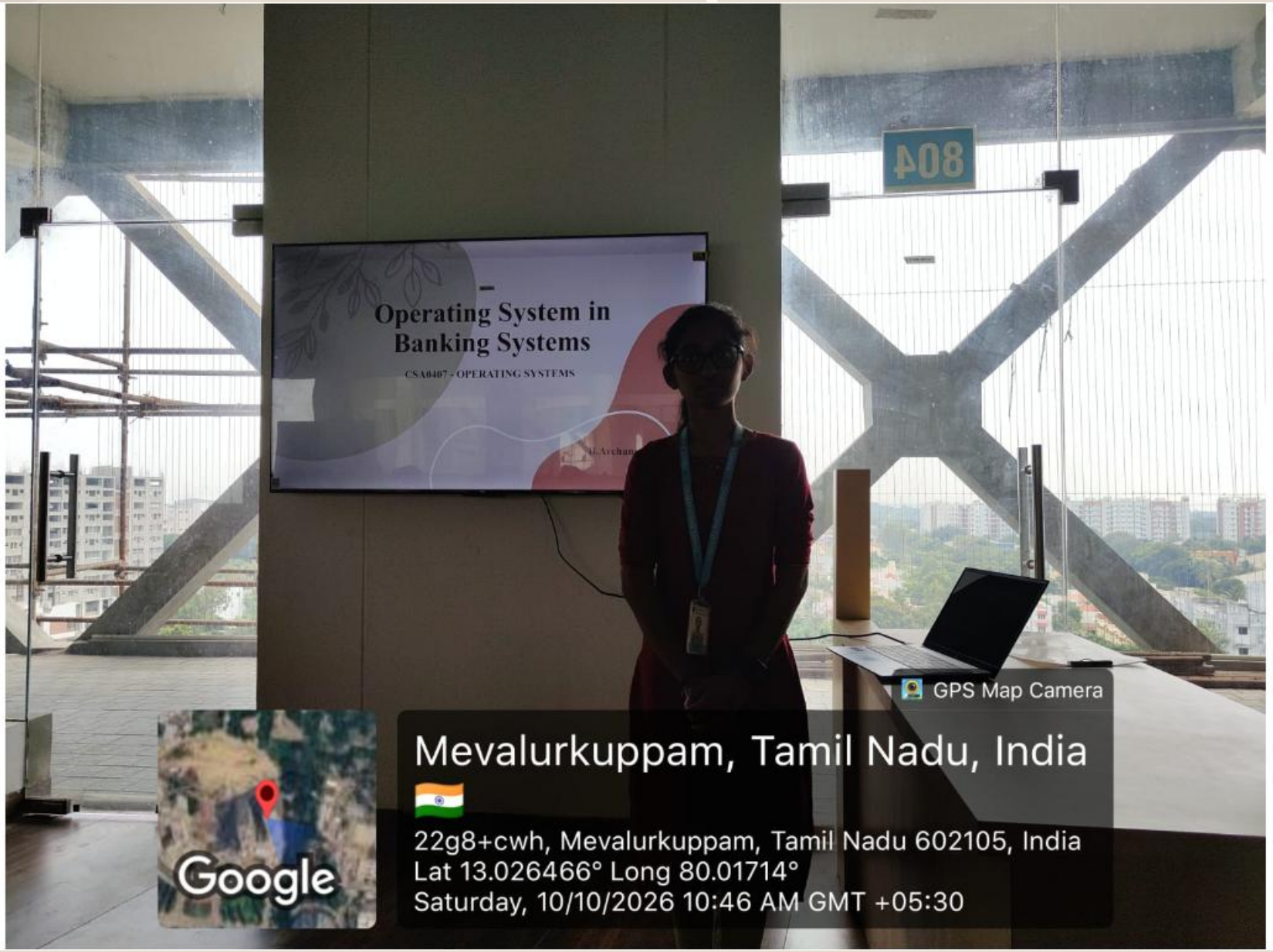
1. Customer enters the transaction details.
2. The banking application checks the details.
3. The OS manages the processing tasks.
4. The banking database updates the account securely.
5. The customer receives a success or failure message.

Benefits:

- Handles many transactions at the same time.
- Protects data from unauthorized access.
- Helps maintain accurate account balances.
- Supports reliable processing and recovery.

Conclusion

- The OS manages thousands of banking transactions simultaneously.
- It supports multiple users and tasks efficiently.
- The OS structure manages hardware and software resources.
- System calls help applications access OS services.
- Security mechanisms protect banking data.
- It helps ensure transactions are fast, secure, accurate, and reliable.


Operating System in
Banking Systems
CSA0407 - OPERATING SYSTEMS
804
GPS Map Camera
Mevalurkuppam, Tamil Nadu, India
22g8+cwh, Mevalurkuppam, Tamil Nadu 602105, India
Lat 13.026466° Long 80.01714°
Saturday, 10/10/2026 10:46 AM GMT +05:30
Google

THANK YOU